Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Gong, Bing Ph.D.		
Project Title:	A New Class of Unnatural Folding Oligomers: Helices with Nano-Sized Cavities		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14260-3000	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	09/01/2000	**End Date:**	09/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
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Performance Goal Text:	n/a		

This research involves the design of a novel class of oligomers that automatically fold into helices with large (10 Å to 30 Å) tubular cavities. Such hollow helices are suggested as potential prototypes of functional nanotubes. The oligomers consist of benzene rings linked by amide groups. The backbone of an oligomer is curved due to the incorporation of intramolecular hydrogen bonds that rigidify the amide linkages. As a result, a backbone that is long enough will fold back on itself, leading to left- and right-handed helical conformations. The helix formed is further stabilized by stacking of the aromatic rings of neighboring spiral turns. Such a backbone-based helical programming will lead to helices whose folded conformation is resilient toward structural variation of the side groups that determine the outside surface properties. The interior of a helix is featured by the amide-O atoms, which makes the tubular cavities rather electronegative and hydrophilic. The internal diameters of the helices are adjustable by combining meta- and para-disubstituted benzene rings or by using larger aromatic rings such as derivatives of naphthalene and anthracene.

Task Description:	These helices, as nanotubes, can be used as artificial pore-forming agents that can be easily functionalized. Molecular gatekeepers based on these nanotubes can be designed by including biochemical, chemical, and physical switches into the structures. The design of pore-forming drugs, drug carriers and sensitive membrane-bound sensors can be envisioned. This research involves the design of a novel class of oligomers that automatically fold into helices with large (10 Å to 30 Å) tubular cavities. These helices, as nanotubes, can be used as artificial pore-forming agents that can be easily functionalized. Molecular gatekeepers based on these nanotubes can be designed by including biochemical, chemical, and physical switches into the structures. The design of pore-forming drugs, drug carriers and sensitive membrane-bound sensors can be envisioned. Hollow helices are suggested as potential prototypes of functional nanotubes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)